

Section 6(a).7 Main PCB Schematic and 6(a).8 PCB Layout

Insert for item 6 "Design and manufacturing information", a. Complete drawings

Item	Details
Board	Main board (motherboard) of the GLD — EasyEDA project "MotherBoardGLDVer2", design package "GLD V4"
Source files	1-Schematic_MotherBoardGLDVer2.json (EasyEDA 6.5.57) and the corresponding PCB layout of the same project
Schematic revision / date	1.0 / 2026-07-19 (from the title block of the EasyEDA schematic)
Drawing numbers	PGLD-GLD-V4-MB-SCH-01 ... -26 (schematic) and PGLD-GLD-V4-MB-PCB-01 ... -10 (layout) — proposed numbering, to be aligned with the applicant's document control
Scope	Main board only. The sensor-module board (8 pcs, connected to H1–H8) is a separate PCB and is not covered by this insert

6(a).7 Main PCB schematic

The electrical schematic of the GLD main board is provided as the actual circuit schematic (not a block diagram). It is presented as 26 figures, each cut from the original EasyEDA schematic around one functional block. Every component in a figure is enclosed in a numbered boundary box, and the same number is used in the description table printed with that figure. Each figure carries a sheet information table with the drawing/document number, revision, date, title and sheet number (n of 26).

Sheet register

Drawing no.	Title	Requirement covered
PGLD-GLD-V4-MB-SCH-01	24 VDC input, input protection and 24 V to 5 V buck (U36)	24 VDC input; input protection
PGLD-GLD-V4-MB-SCH-02	Power connector J1, battery input and external 5 V	24 VDC input
PGLD-GLD-V4-MB-SCH-03	Battery load switch (U14)	Input protection
PGLD-GLD-V4-MB-SCH-04	5 V power-path mux (U13)	5 V rail
PGLD-GLD-V4-MB-SCH-05	Battery boost to about 24 V (U1, TPS61175)	5 V rail / +24 V backup
PGLD-GLD-V4-MB-SCH-06	Battery boost to 5 V (U41, TPS61088)	5 V rail
PGLD-GLD-V4-MB-SCH-07	5 V to 3.3 V buck (U43)	3.3 V rail
PGLD-GLD-V4-MB-SCH-08	Always-on 3.3 V LDO and its source selection (U42)	3.3 V rail
PGLD-GLD-V4-MB-SCH-09	TPL5010 timer/watchdog (U46)	3.3 V rail
PGLD-GLD-V4-MB-SCH-10	ENA latch (U51) and open-drain inverter (U15)	3.3 V rail
PGLD-GLD-V4-MB-SCH-11	3V3AON voltage supervisor (U16)	3.3 V rail
PGLD-GLD-V4-MB-SCH-12	Sensor module connector H2 (one connector as example)	MQ heater supply
PGLD-GLD-V4-MB-SCH-13	Sensor module connectors H1–H8	MQ heater supply; sensor interface
PGLD-GLD-V4-MB-SCH-14	Sensor module enable I/O expander (U12)	MQ heater supply (module enable)
PGLD-GLD-V4-MB-SCH-15	VMID bias of about 2.5 V (U3)	MQ sensor analog circuitry
PGLD-GLD-V4-MB-SCH-16	Analog +5VA supply filter (L6)	MQ sensor analog circuitry
PGLD-GLD-V4-MB-SCH-17	Sensor module I2C multiplexer (U33)	MQ sensor analog circuitry
PGLD-GLD-V4-MB-SCH-18	Temperature and humidity sensor (U44, SHT40)	Analog circuitry (environment)
PGLD-GLD-V4-MB-SCH-19	ADS1256 ADC with input filters (AIN), clock and decoupling	ADC; analog circuitry
PGLD-GLD-V4-MB-SCH-20	2.5 V voltage reference and VREF buffer	ADC
PGLD-GLD-V4-MB-SCH-21	ESP32-S3-WROOM-1U-N16R8 with support parts (incl. status LED)	ESP32-S3; LED
PGLD-GLD-V4-MB-SCH-22	USB, CH340C and auto-reset	ESP32-S3 (programming)
PGLD-GLD-V4-MB-SCH-23	E22-900MM22S LoRa module and antenna connector	LoRa
PGLD-GLD-V4-MB-SCH-24	5 V fan driver (Q5, J3)	Fan driver
PGLD-GLD-V4-MB-SCH-25	24 V alarm output driver (Q4, J2)	LED / buzzer (alarm) driver

Drawing no.	Title	Requirement covered
PGLD-GLD-V4-MB-SCH-26	RS-485 transceiver (U47)	RS-485 / Modbus

All sheets: revision 1.0, date 2026-07-19.

Coverage of the required schematic content

Required content (6(a).7)	Where shown	Status
24 VDC input	Sheets 01, 02	Provided
Input protection	Sheets 01–03; protection-device table below	Provided — see findings 1–2
5 V rail	Sheets 01 (buck U36), 04–06	Provided
3.3 V rail	Sheets 07–11	Provided
MQ heater supply	Sheets 12–14 (+5 V and enable to each sensor module)	Partly — heater switch is on the sensor module (finding 4)
MQ sensor bridge / analog circuitry	Sheets 15–18; AIN filters on sheet 19	Partly — load resistor is on the sensor module (finding 4)
ADC	Sheets 19, 20	Provided
ESP32-S3	Sheets 21, 22	Provided
LoRa	Sheet 23	Provided
Fan driver	Sheet 24	Provided
LED / buzzer driver	Sheet 25 (24 V alarm output J2); status LED on sheet 21	Provided
RS-485 / Modbus	Sheet 26	Provided
Protection devices	Table below	Provided
Grounding	Text below	Provided — PE point not drawn (finding 5)
Title block per sheet	Sheet information table on every figure	Provided

Protection devices

Designator	Part	Location	Function	Sheet
F1	MINISMDC260F/16-2	Series with 24V+ after J1	Overcurrent on 24 V line (PTC)	01
D1	SMBJ33A	After F1, to 24V–	Surge / transient clamp on 24 V line (TVS)	01
L1	ACM7060-301-2PL	Series with 24V+ and return	Common-mode noise	01
L4	MPZ2012S101AT000	Series with +24V	High-frequency noise (ferrite)	01
D2 / R26	LBZT52C12T1G / 100k	Gate of Q3	12 V gate-source limit for Q3	01
Q3	SI7465DP-T1-GE3	+24V to buck input	High-side switch	01
F2	MINISMDC260F/16-2	Series with battery input J1.2	Battery overcurrent (PTC)	02
Q2 / R1	AO4407 / 100k	Battery to VBAT_IN	High-side switch	02
D7 / D13	SS54	5VEXT to +5V; boost to +24V	Reverse-current blocking (OR-ing)	02, 05
D3 / D10	BAT54C	Source of U42 (3V3AON)	OR-ing of supply sources	08
D8 / D9	SS14	Loads J2 and J3	Flyback for inductive loads	24, 25
D4 / D5 / D11	LES5D5.0CT1G	USB D–, D+, VBUS	USB ESD	22
D6	SM712	RS-485 A/B	RS-485 line TVS	26
L6	GZ1005D600TF	+5V to +5VA	Analog supply noise isolation	16

Grounding

- The circuit uses a single GND net; there is no separate AGND or chassis-ground net. The AGND/PGND/DGND pins of all ICs are connected to GND.
- The 24 V return (J1 pin 5, 24V–) enters GND through one winding of common-mode choke L1 (pin 2 to pin 3). J1 pins 1 and 3 are connected directly to GND.
- The analog supply +5VA is separated from +5V by ferrite bead L6; analog and digital ground are joined at GND.
- The shell/EP pads of USB1 and the GND pads of the LoRa module/antenna connector are connected to GND.

- A connection point from circuit GND to the enclosure / protective earth is not drawn on the schematic; it is defined by the enclosure grounding drawing (6(a).10).

Findings to be resolved before the dossier is finalised

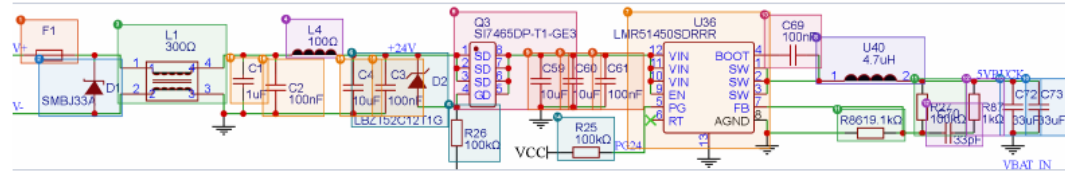
#	Finding (from the schematic)	Action
1	F1 and F2 are MINISMDC260F/16-2; the "/16" suffix normally means a 16 V rating, while F1 is on the 24 V line.	Confirm the PTC voltage rating from the datasheet or change the part.
2	Q3 (24 V) and Q2 (battery) have the source on the input side and the gate pulled to GND; with this orientation the body diode does not block reverse polarity.	Confirm the intended function. The dossier text "P-channel MOSFET reverse-polarity protection" (§3.6) must match the actual circuit.
3	Battery input, battery boost to 24 V (TPS61175) and to 5 V (TPS61088), battery load switch and power-path mux are populated on the main board.	Decide whether the battery path is excluded (depopulated / not fitted) or included in the certified configuration; the production configuration is stated as 24 VDC only.
4	Each sensor connector H1–H8 carries only +5 V, VMID, enable, I2C and AIN. The per-channel heater switch and the sensor load resistor are not on the main board.	Provide the sensor-module schematic and layout as a separate drawing set.
5	Single GND net; no chassis/PE connection point on the board.	Show the GND-to-enclosure/PE arrangement in the grounding drawing (6(a).10).
6	RS-485 has TVS D6 only; no 120 Ω termination or A/B bias resistors.	Confirm termination/bias is provided externally (installation note) or is not required.
7	Converter output voltages (≈ 5.0 V, ≈ 24 V, ≈ 5.3 V) are calculated from feedback resistors, not measured.	Record measured rail voltages in the routine/functional test.
8	The load type on alarm output J2 (24 V siren/beacon) is not defined on the schematic.	Identify the alarm module (manufacturer/model, current) in the BOM (6.b).

Schematic figures (sheets 01–26)

The pages below reproduce the schematic figures exactly as issued, including the per-figure sheet information table and component description table.

24 VDC input

Figure 1 - 24 VDC input, input protection and 24 V to 5 V buck (U36)

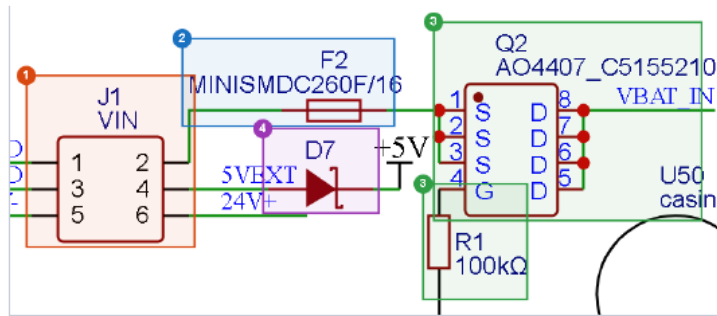


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-01	1.0	2026-07-19	24 VDC input, input protection and 24 V to 5 V buck (U36)	1 of 26

Block function: The 24 VDC path from the connector to the 24 V to 5 V buck (U36). **Requirement covered:** 24 VDC input and input protection

No	Component	What it is	What it does
1	F1	Resettable PTC fuse MINISMDC260F/16-2 (2.6 A hold).	Opens on a short circuit or overload on the 24 V line and recovers by itself. Note: the /16 suffix in the part number usually means a 16 V voltage rating - to be confirmed against the datasheet because this line is 24 V.
2	D1	TVS diode SMBJ33A (33 V standoff).	Clips voltage surges/transients between the 24 V line after F1 and 24V-, so the circuitry behind it is not damaged.
3	L1	Common-mode choke ACM7060-301-2PL (4 pins).	Suppresses common-mode noise on the 24V+ / 24V- pair. Winding 1-4 carries 24V+, winding 2-3 carries the return side (24V- to GND).
4	L4	Ferrite bead MPZ2012S101A (100 ohm @ 100 MHz).	High-frequency noise filter in series with the +24V line before it enters the internal rail.
5	Q3	P-channel MOSFET SI7465DP (source on +24V, drain toward the buck input).	Passes +24V to the 5 V converter. Its source is on the +24V side and its drain on the load side; this orientation should be verified if the part is meant to provide reverse-polarity protection.
6	D2, R26	12 V zener (LBZT52C12) and a 100k resistor to GND.	R26 pulls the gate to GND so Q3 turns on. D2 limits the gate-source voltage of Q3 to 12 V so the gate is not damaged by 24 V.
7	U36	Synchronous buck IC LMR51450 (VIN up to 36 V).	Step-down converter from 24 V to 5VBUCK. EN is tied to VIN (always on); the PG (power good) pin goes to the ESP32 as signal PG24.
8	U40	4.7 uH inductor (GSSM06304R7M2AU).	Buck energy-storage inductor between the SW node and 5VBUCK.
9	C59, C60, C61	Input capacitors 10 uF/50 V x2 + 100 nF.	Supply the pulsed input current of the buck.
10	C72, C73	Output capacitors 33 uF x2 + 10 uF.	Smooth the 5VBUCK output voltage.
11	R27, R86	FB voltage divider 100k / 19.1k.	Sets the output: $0.8 \text{ V} \times (1 + 100/19.1) = \text{about } 5.0 \text{ V}$ (calculated from the resistor values).
12	R87, C70	1k resistor in series with a 33 pF capacitor on FB.	Feed-forward network for stability / transient response.
13	C69	100 nF BOOT-SW capacitor.	Bootstrap for the internal high-side MOSFET driver.
14	R25	100k pull-up to VCC on PG24.	Gives a 3.3 V logic level for the power-good signal read by the ESP32 (IO47).
15	C1, C2, C3, C4	Decoupling capacitors 1 uF, 100 nF, 100 nF, 10 uF/50 V.	Filter ripple/noise and hold up small transients on the +24V rail (paired with L1/L4).

Figure 2 - Power connector J1, battery input and external 5 V



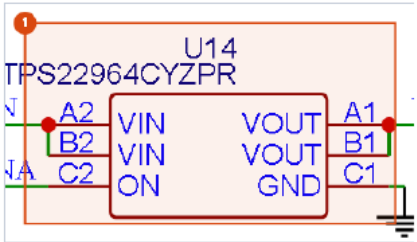
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-02	1.0	2026-07-19	Power connector J1, battery input and external 5 V	2 of 26

Block function: Power connector, battery input protection and switch, and the external 5 V path. **Requirement covered:** 24 VDC input and input protection

No	Component	What it is	What it does
1	J1	Power connector, 2x3 pins (J1).	The only power entry from outside: pin 6 = 24V+, pin 5 = 24V-, pin 4 = 5VEXT (external 5 V), pins 3 and 1 = GND, pin 2 = battery input (goes to F2).
2	F2	PTC fuse MINISMDC260F/16-2.	Overcurrent protection for the battery input at J1 pin 2.
3	Q2, R1	P-MOSFET AO4407 with a 100k gate resistor to GND.	Passes the battery to net VBAT_IN. R1 turns Q2 on through the gate to GND. The source-on-battery orientation should be verified if the part is meant to provide reverse-polarity protection.
4	D7	SS54 Schottky diode from 5VEXT to +5V.	Lets external 5 V (J1 pin 4) supply the +5V rail without back-feeding.

Input protection

Figure 3 - Battery load switch (U14)



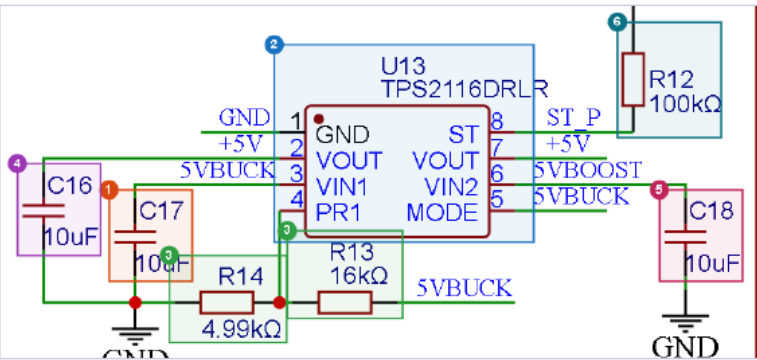
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-03	1.0	2026-07-19	Battery load switch (U14)	3 of 26

Block function: Battery load switch to the VBAT rail. **Requirement covered:** Input protection

No	Component	What it is	What it does
1	U14	Load switch TPS22964C (VIN = VBAT_IN, VOUT = VBAT, ON = ENA).	Connects/disconnects the battery to the whole VBAT rail (boost converters) according to the ENAsignal from the power-latch circuit.

5 V rail

Figure 4 - 5 V power-path mux (U13)

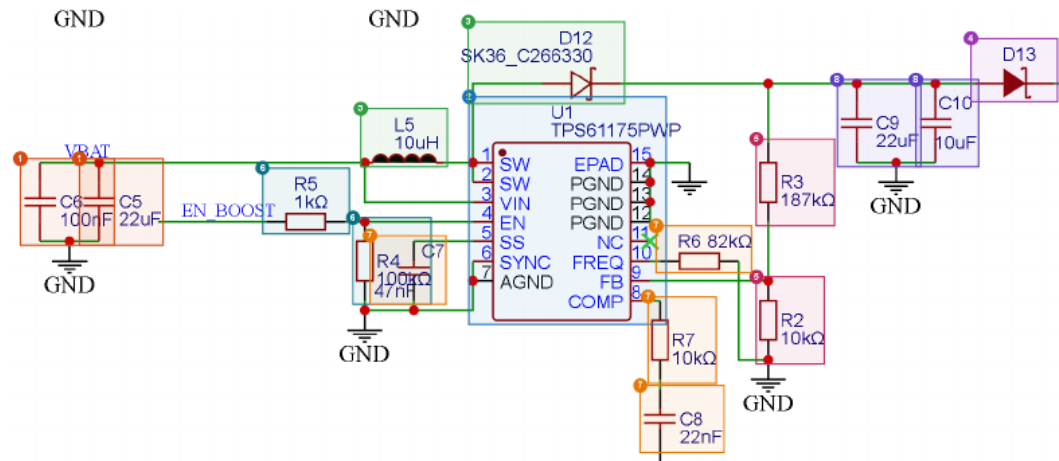


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-04	1.0	2026-07-19	5 V power-path mux (U13)	4 of 26

Block function: Selects the 5 V source between 5VBUCK and 5VBOOST. **Requirement covered:** 5 V rail

No	Component	What it is	What it does
1	C17	Output capacitors 33 uF x2 + 10 uF.	Smooth the 5VBUCK output voltage.
2	U13	Power mux TPS2116 (VIN1 = 5VBUCK, VIN2 = 5VBOOST, VOUT = +5V).	Automatically selects the 5 V source with priority on 5VBUCK (MODE is tied to 5VBUCK) and switches to 5VBOOST when 5VBUCK is lost. The ST pin goes to the ESP32 as ST_P (source status).
3	R13, R14	16k / 4.99k divider on PR1.	Sets the voltage threshold at which 5VBUCK is considered valid before the mux gives it priority.
4	C16	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
5	C18	VBAT input and 5VBOOST output capacitors.	Decoupling on the input and output sides of the boost.
6	R12	100k pull-up to VCC on ST_P.	Gives a 3.3 V logic level for the mux status signal read by the ESP32 (IO18).

Figure 5 - Battery boost to about 24 V (U1, TPS61175)

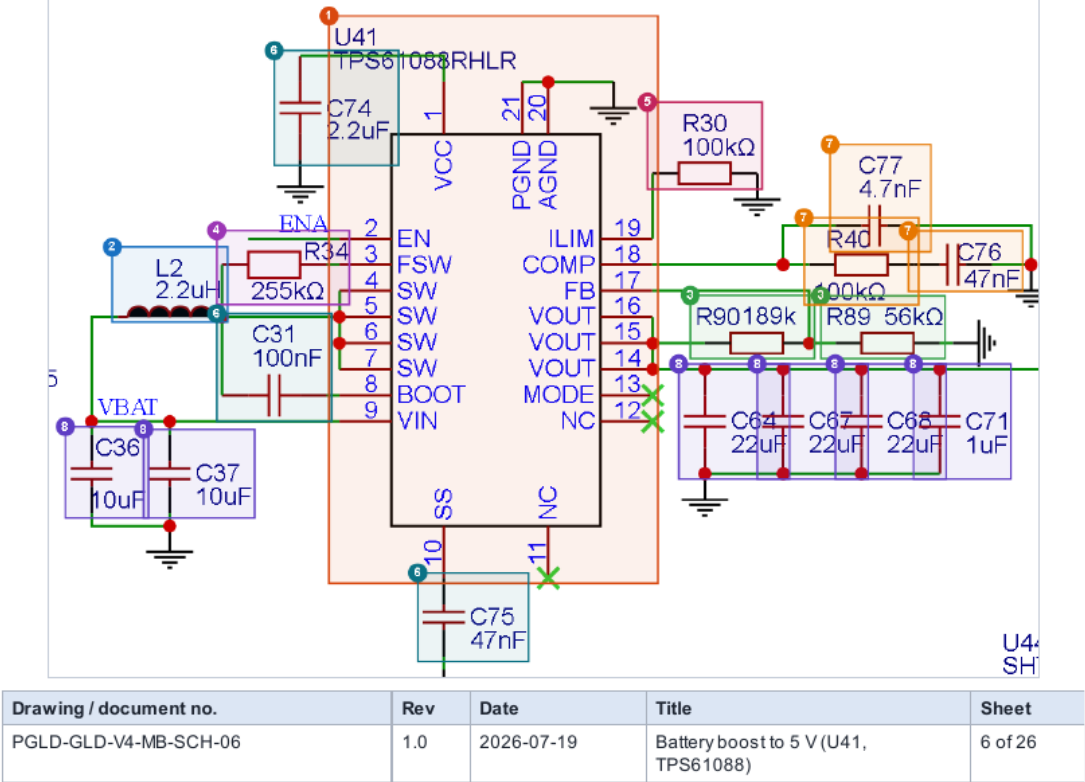


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-05	1.0	2026-07-19	Battery boost to about 24 V (U1, TPS61175)	5 of 26

Block function: Boosts the battery to about 24 V as a backup of the +24V rail. **Requirement covered:** 5 V rail / backup of the +24V rail

No	Component	What it is	What it does
1	C5, C6	22 uF and 100 nF capacitors on VBAT.	Decoupling on the load-switch output.
2	U1	Boost converter TPS61175 (VIN = VBAT).	Boosts the battery voltage to about 24 V. Enabled by the EN_BOOST signal from the ESP32 (IO15).
3	L5, D12	10 uH inductor and SK36 Schottky diode.	Boost inductor and rectifier diode from the SW node to the boost output.
4	D13	SS54 Schottky diode from the boost output to +24V.	ORs the boost output onto the +24V rail without back-feeding into the boost converter.
5	R3, R2	FB divider 187k / 10k.	Sets the output to about $1.229 \text{ V} \times (1 + 187/10) = 24 \text{ V}$ (calculated from the resistor values).
6	R4, R5	100k pull-down and 1k series resistor on EN.	Keep the boost off while EN_BOOST is inactive; the 1k limits current from the GPIO.
7	R6, R7, C7, C8	82k FREQ resistor, 10k/22 nF COMP network, 47 nF soft-start.	Set the switching frequency, loop compensation and soft-start.
8	C9, C10	Output capacitors 22 uF/35 V and 10 uF/50 V.	Boost output filter.

Figure 6 - Battery boost to 5 V (U41, TPS61088)

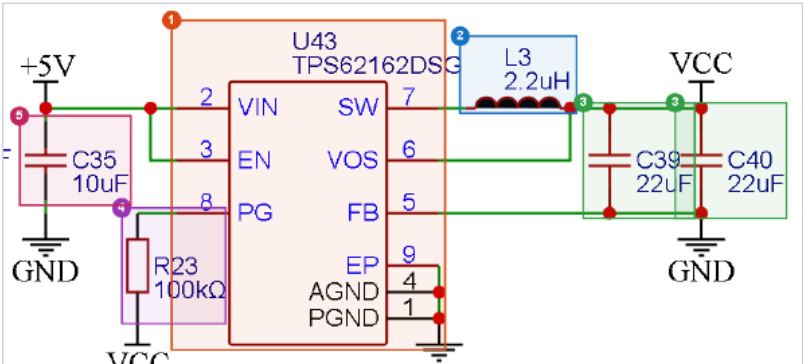


Block function: Boosts the battery to 5VBOOST. Requirement covered: 5 V rail

No	Component	What it is	What it does
1	U41	Boost converter TPS61088 (VIN = VBAT, VOUT = 5VBOOST, EN = ENA).	Generates 5 V from the battery. Active when the ENAsignal from the power-latch circuit is high.
2	L2	2.2 uH inductor.	Boost inductor between VBAT and the SW node.
3	R90, R89	FB divider 189k / 56k.	Sets the output to about 1.204 V x (1 + 189/56) = 5.3 V (calculated from the resistor values).
4	R34	255k resistor on FSW.	Sets the switching frequency.
5	R30	100k resistor on ILIM.	Sets the peak current limit.
6	C74, C75, C31	Internal VCC capacitor, soft-start capacitor and bootstrap capacitor.	Support parts for the IC.
7	R40, C76, C77	COMP compensation network.	Stabilizes the boost control loop.
8	C64, C67, C68, C71, C36, C37	VBAT input and 5VBOOST output capacitors.	Decoupling on the input and output sides of the boost.

3.3 V rail

Figure 7 - 5 V to 3.3 V buck (U43)

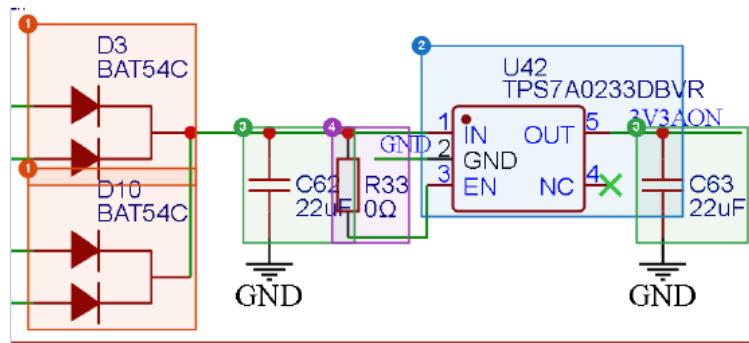


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-07	1.0	2026-07-19	5 V to 3.3 V buck (U43)	7 of 26

Block function: Buck that generates the 3.3 V VCC rail. **Requirement covered:** 3.3 V rail

No	Component	What it is	What it does
1	U43	Buck TPS62162 (VIN = +5V, EN = +5V, VOS = VCC).	Steps +5V down to 3.3 V (fixed-voltage version, FB to GND). Always on because EN is tied to +5V.
2	L3	2.2 uH inductor.	Buck inductor from the SW node to VCC.
3	C39, C40	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
4	R23	100k pull-up on PG.	PG is not connected to any other component in this netlist (pull-up only).
5	C35	+5V capacitor, PG pull-up and VCC output capacitors.	Decoupling of the input/output of the 3.3 V buck; R23 is the pull-up on the PG pin.

Figure 8 - Always-on 3.3 V LDO and its source selection (U42)

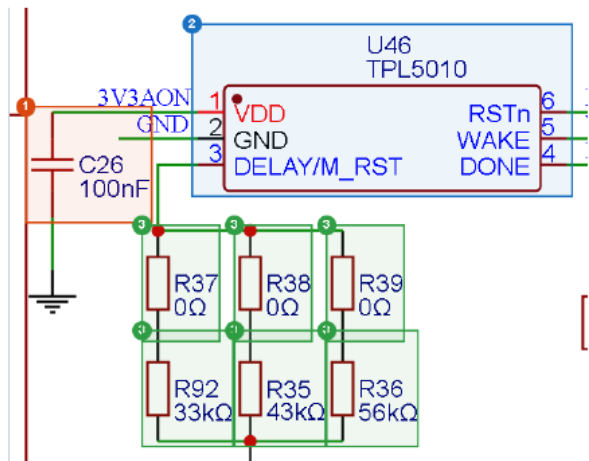


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-08	1.0	2026-07-19	Always-on 3.3 V LDO and its source selection (U42)	8 of 26

Block function: The 3V3AON LDO and its source selection. **Requirement covered:** 3.3 V rail

No	Component	What it is	What it does
1	D3, D10	Dual diode BAT54C (common cathode).	D3 ORs VBAT_IN and 5VBUCK; D10 ORs 5VEXT and VBUS (USB). The result is the net that feeds the LDO input.
2	U42	LDO TPS7A0233 (3.3 V, very low quiescent current).	Converts the available source into 3V3AON. EN is pulled to the input through R33 (0 ohm), so it is always enabled.
3	C62, C63	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
4	R33	0 ohm resistor between the input and the EN pin of U42.	Ties EN to the input so the LDO is always enabled (can be removed if a separate EN is needed).

Figure 9 - TPL5010 timer/watchdog (U46)

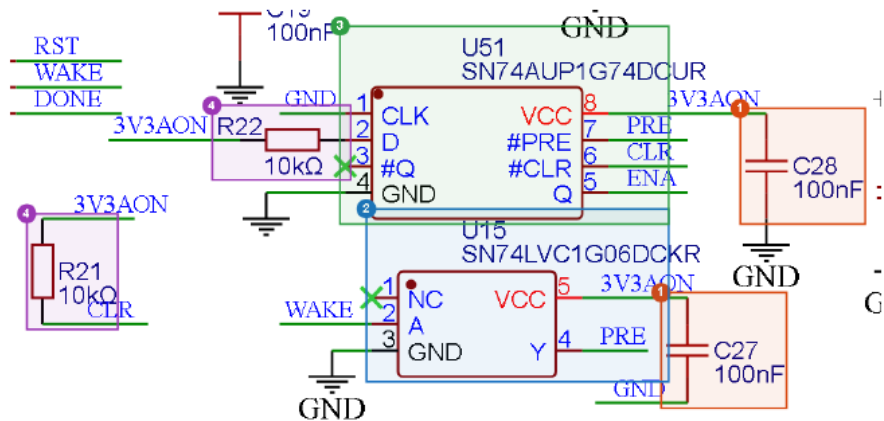


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-09	1.0	2026-07-19	TPL5010 timer/watchdog (U46)	9 of 26

Block function: Nano-power timer/watchdog and its interval selector. **Requirement covered:** 3.3 V rail / power latch

No	Component	What it is	What it does
1	C26	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U46	Nano-power timer TPL5010.	Generates periodic WAKE pulses and works as a watchdog: DONE from the ESP32 (IO17) signals that the firmware is still alive, RSTn goes to RST (ESP32 reset).
3	R35, R36, R92, R37, R38, R39	43k / 56k / 33k resistors to GND with three 0 ohm jumpers.	Select the TPL5010 timer interval through the DELAY/M_RST pin (one 0 ohm jumper is fitted for the chosen interval).

Figure 10 - ENAlatch (U51) and open-drain inverter (U15)

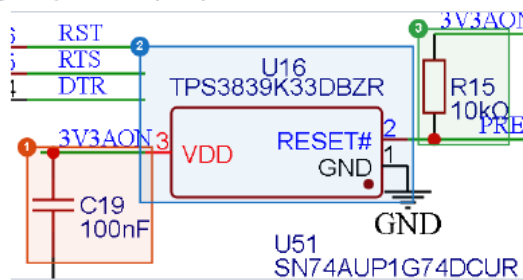


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-10	1.0	2026-07-19	ENAlatch (U51) and open-drain inverter (U15)	10 of 26

Block function: ENAlatch flip-flop and open-drain inverter. Requirement covered: 3.3 V rail / power latch

No	Component	What it is	What it does
1	C27, C28	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U15	Open-drain inverter SN74LVC1G06.	Turns WAKE from U46 into a pull-down on net PRE (R15 pull-up to 3V3AON).
3	U51	D flip-flop SN74AUP1G74 (CLK to GND, Q = ENA).	Used as a latch: PRE low sets ENA, CLR low from the ESP32 (IO16) clears ENA. ENA controls U14 (load switch) and U41 (5 V boost).
4	R21, R22	10k pull-ups to 3V3AON on PRE, CLR and the flip-flop D input.	Keep the default logic levels.

Figure 11 - 3V3AON voltage supervisor (U16)



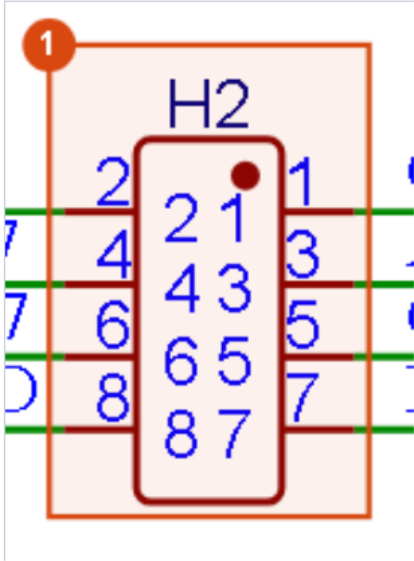
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-11	1.0	2026-07-19	3V3AON voltage supervisor (U16)	11 of 26

Block function: Voltage supervisor of 3V3AON. Requirement covered: 3.3 V rail / power latch

No	Component	What it is	What it does
1	C19	22 uF and 100 nF input/output capacitors.	Decoupling of the LDO and the 3V3AON rail.
2	U16	Voltage supervisor TPS3839K33 (threshold variant for a 3.3 V rail).	RESET# output is connected to net PRE; supervises 3V3AON.
3	R15	10k pull-ups to 3V3AON on PRE, CLR and the flip-flop D input.	Keep the default logic levels.

MQ heater supply

Figure 12 - Sensor module connector H2 (one connector as example)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-12	1.0	2026-07-19	Sensor module connector H2 (one connector as example)	12 of 26

Block function: One sensor module connector as an example of every pin function. **Requirement covered:** MQ heater supply and MQ sensor interface

No	Component	What it is	What it does
1	H2	Sensor module connectors (H1..H8).	Each connector carries: pin 2 = +5V (module/heater supply), pin 8 = VMID (bias), pin 3 = AINx (analog signal to the ADC), pins 4/6 = SCLx/SDAx (I2C channel from the mux), pin 7 = ENx (module enable), pins 1/5 = GND. The heater switch and sensor load resistor are not on this board (assumed to be on the sensor module; to be confirmed).

Figure 13 - Sensor module connectors H1-H8

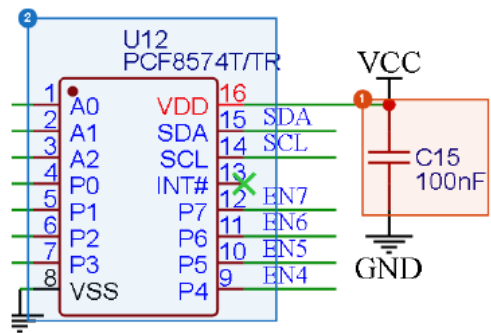


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-13	1.0	2026-07-19	Sensor module connectors H1-H8	13 of 26

Block function: The eight sensor module connectors H1-H8. **Requirement covered:** MQ heater supply and MQ sensor interface

No	Component	What it is	What it does
1	H1, H2, H3, H4, H5, H6, H7, H8	Sensor module connectors (H1..H8).	Each connector carries: pin 2 = +5V (module/heater supply), pin 8 = VMID (bias), pin 3 = AINx (analog signal to the ADC), pins 4/6 = SCLx/SDAx (I2C channel from the mux), pin 7 = ENx (module enable), pins 1/5 = GND. The heater switch and sensor load resistor are not on this board (assumed to be on the sensor module; to be confirmed).

Figure 14 - Sensor module enable I/O expander (U12)



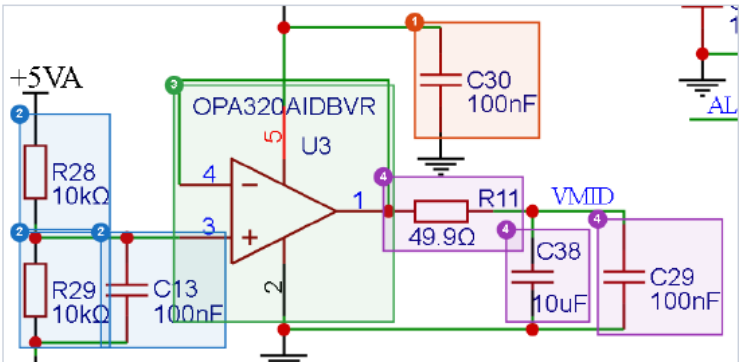
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-14	1.0	2026-07-19	Sensor module enable I/O expander (U12)	14 of 26

Block function: I/O expander that enables the sensor modules. **Requirement covered:** MQ heater supply (module enable)

No	Component	What it is	What it does
1	C15	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U12	I/O expander PCF8574.	Outputs P0-P7 become EN0-EN7 to enable the sensor modules one by one.

MQ sensor bridge / analog circuitry

Figure 15 - VMID bias of about 2.5 V (U3)

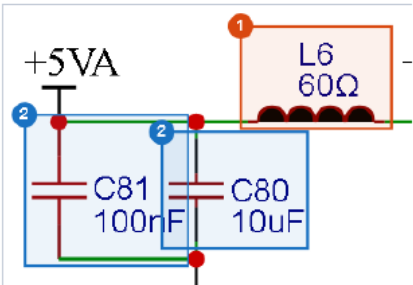


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-15	1.0	2026-07-19	VMID bias of about 2.5 V (U3)	15 of 26

Block function: Generator of the VMID bias of about 2.5 V. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C30	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
2	R28, R29, C13	10k / 10k divider and 100 nF.	Creates half of +5VA (about 2.5 V).
3	U3	Op-amp OPA320 as a buffer.	Buffers the mid-voltage into a low-impedance VMID source.
4	R11, C29, C38	49.9 ohm resistor and capacitors.	Isolate the buffer output and keep it stable with capacitive loads; VMID goes to pin 8 of every sensor connector.

Figure 16 - Analog +5VA supply filter (L6)

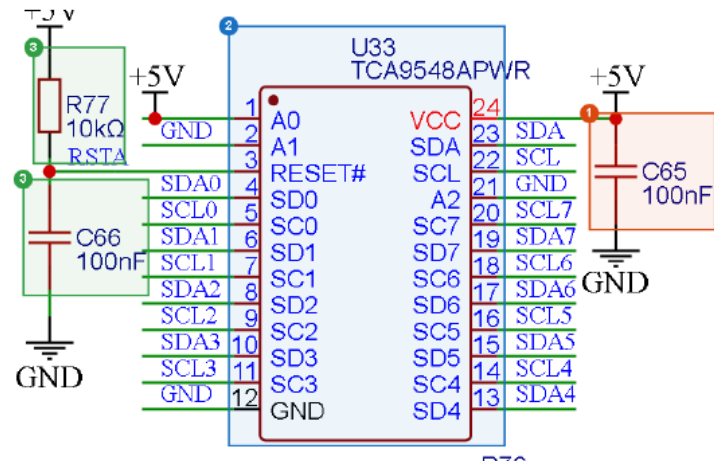


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-16	1.0	2026-07-19	Analog +5VA supply filter (L6)	16 of 26

Block function: Analog +5VA supply filter. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	L6	60 ohm ferrite bead.	Separates digital noise from the analog +5VA rail.
2	C80, C81	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.

Figure 17 - Sensor module I2C multiplexer (U33)

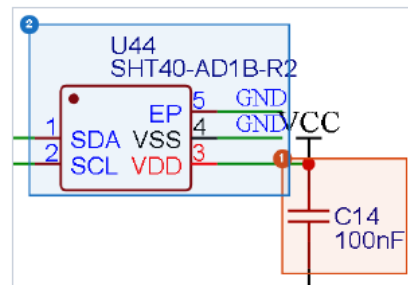


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-17	1.0	2026-07-19	Sensor module I2C multiplexer (U33)	17 of 26

Block function: I2C multiplexer for the eight sensor modules. **Requirement covered:** MQ sensor bridge / analog circuitry

No	Component	What it is	What it does
1	C65	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
2	U33	I2C multiplexer TCA9548A(8 channels).	Splits the main I2C bus into eight channels SDA0-7/SCL0-7 toward the modules; RESET# is held high by R77.
3	R77, C66	10k pull-up and 100 nF on RSTA.	Keep the mux from being reset unless intended.

Figure 18 - Temperature and humidity sensor (U44)



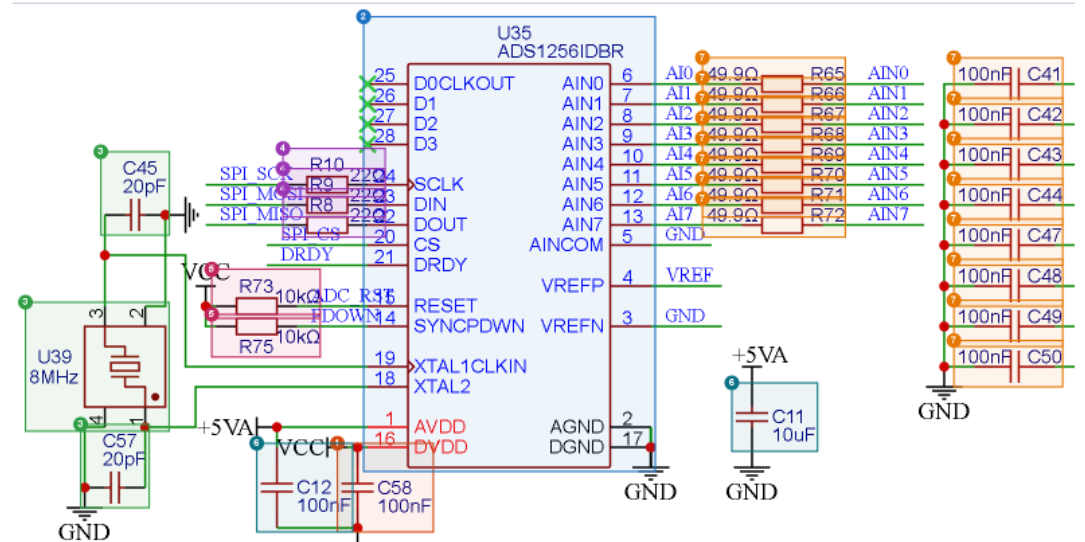
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-18	1.0	2026-07-19	Temperature and humidity sensor (U44)	18 of 26

Block function: Environment sensor inside the board. **Requirement covered:** MQ sensor bridge / analog circuitry (environment compensation)

No	Component	What it is	What it does
1	C14	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U44	Temperature/humidity sensor SHT40.	Measures temperature and humidity inside the board on the main I2C bus.

ADC

Figure 19 - ADS1256 ADC with input filters (AIN), clock and decoupling



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-19	1.0	2026-07-19	ADS1256 ADC with input filters (AIN), clock and decoupling	19 of 26

Block function: ADS1256 with its input filters, clock and decoupling. **Requirement covered:** ADC; MQ sensor bridge / analog circuitry

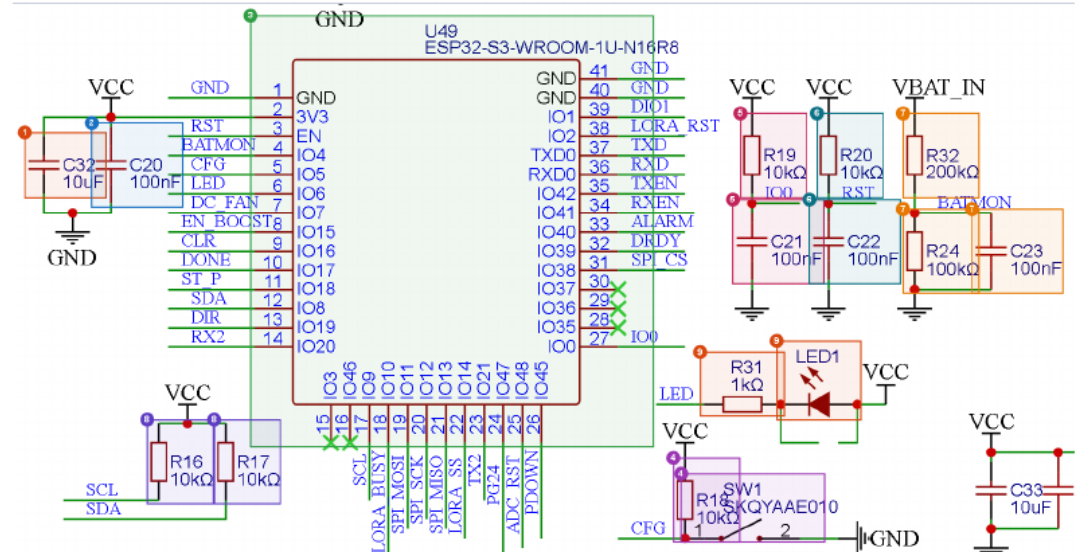
No	Component	What it is	What it does
1	C58	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U35	Delta-sigma ADC ADS1256 (8 channels, 24 bit).	Converts the AI0-AI7 voltages to digital data. AVDD = +5VA, DVDD = VCC, VREFP = VREF, VREFN and AINCOM = GND. Controlled over SPI (CS, SCLK, DIN, DOUT) + DRDY.
3	U39, C45, C57	4-pin SMD crystal with two 20 pF capacitors.	ADC clock source (8 MHz according to the U39 label on the schematic).
4	R8, R9, R10	22 ohm series resistors on MISO, MOSI, SCK.	Damp ringing/EMI on the SPI lines to the ADC.
5	R73, R75	10k pull-ups on RESET and SYNC/PDWN.	Keep the ADC active by default; the ESP32 can pull them low (ADC_RST, PDOWN).
6	C11, C12	10 uF and 100 nF capacitors on +5VA.	Decoupling for the ADC, op-amps and reference.
7	R65, R66, R67, R68, R69, R70, R71, R72, C41, C42, C43, C44, C47, C48, C49, C50	49.9 ohm series resistor and 100 nF capacitor to GND on every channel.	About 32 kHz low-pass (calculated) that suppresses noise and protects the ADC inputs; AINx from the connector becomes AIx at the ADS1256.

Block function: 2.5 V reference and VREF buffer. **Requirement covered:** ADC

GLD V2 — Section 6(a).7 Main PCB Schematic / 6(a).8 PCB Layout · Page 20

ESP32-S3

Figure 21 - ESP32-S3-WROOM-1U-N16R8 with support parts (including status LED)

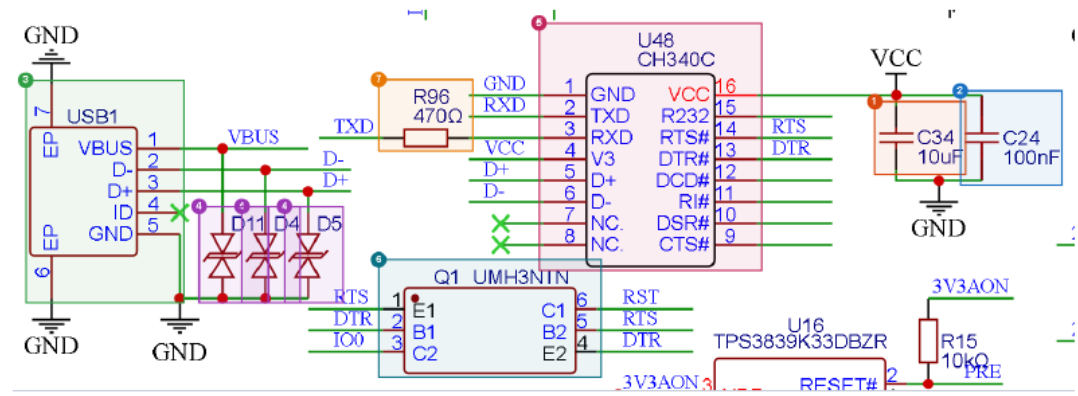


Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-21	1.0	2026-07-19	ESP32-S3-WROOM-1U-N16R8 with support parts (including status LED)	21 of 26

Block function: Main microcontroller with reset, boot, LED and I2C pull-ups. **Requirement covered:** ESP32-S3; LED / buzzer driver (LED)

No	Component	What it is	What it does
1	C32	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C20	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	U49	Module ESP32-S3-WROOM-1U-N16R8 (external antenna, 16 MB flash, 8 MB PSRAM).	The brain of the GLD: reads the ADC (SPI_CS/DRDY/ADC_RST/PDOWN), controls the LoRa radio (LORA_SS/RST/BUSY/DIO1/TXEN/RXEN), I2C (SDA/SCL), RS-485 (TX2/RX2/DIR), alarm, fan, LED and boost, and reads BATMON, PG24, ST_P and CFG.
4	SW1, R18	CFG button with a 10k pull-up.	Configuration button: pulls CFG to GND when pressed.
5	R19, C21	10k pull-up and 100 nF on IO0.	Keep IO0 high (normal boot); Q1 pulls it low in download mode.
6	R20, C22	10k pull-up and 100 nF on EN/RST.	RC reset circuit for the ESP32; also driven by the TPL5010.
7	R32, R24, C23	200k / 100k divider to BATMON with 100 nF.	Scales VBAT_IN down so the ESP32 ADC (IO4) can read it safely; the 1/3 ratio matches the firmware formula 'avg x 3'.
8	R16, R17	10k I2C pull-ups to VCC.	Pull-ups for SDA and SCL.
9	LED1, R31	0603 LED (19-217/BHC-ZL1M2RY/3T) with a 1k resistor.	Lights when the ESP32 pulls the LED pin (IO6) to GND; anode to VCC (active-low).

Figure 22 - USB, CH340C and auto-reset



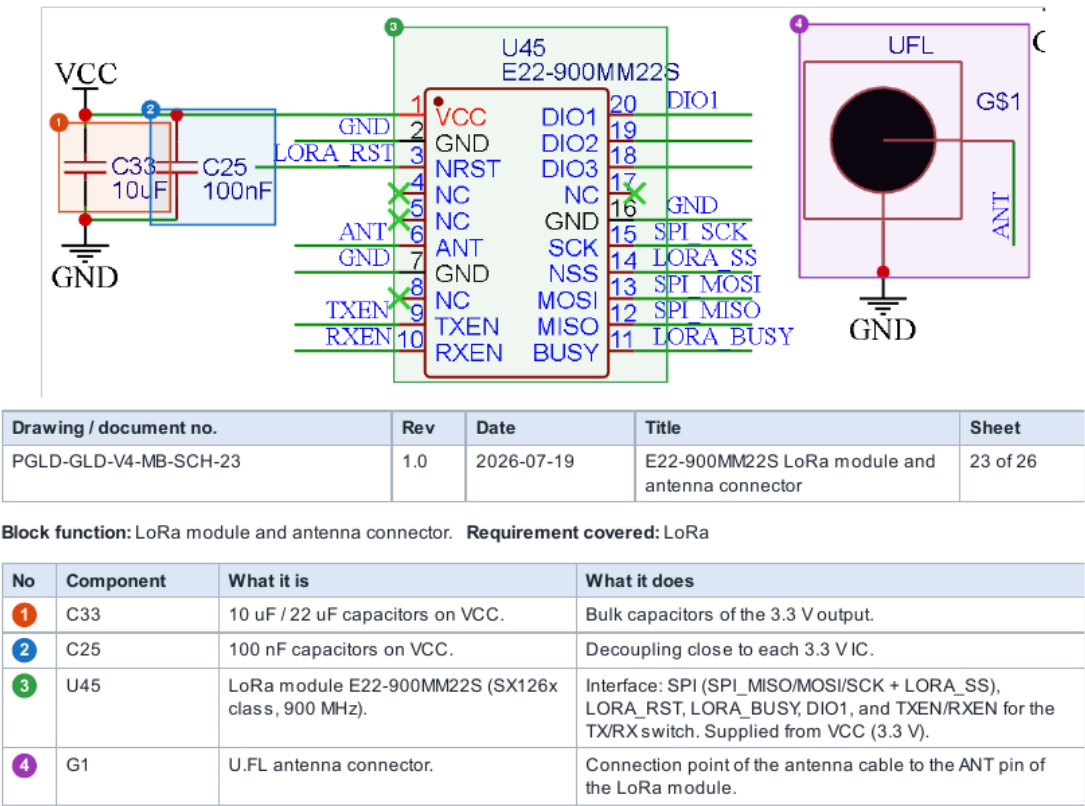
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-22	1.0	2026-07-19	USB, CH340C and auto-reset	22 of 26

Block function: USB connector, ESD protection, CH340C and auto-reset. **Requirement covered:** ESP32-S3 (programming)

No	Component	What it is	What it does
1	C34	10 uF / 22 uF capacitors on VCC.	Bulk capacitors of the 3.3 V output.
2	C24	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
3	USB1	Micro-USB connector.	Programming and serial port to the PC; VBUS also feeds the 3V3AON source OR-ing (3.3 V rail).
4	D4, D5, D11	ESD protection on D-, D+ and VBUS.	Protects the USB lines from electrostatic discharge.
5	U48	USB-UART bridge CH340C (VCC 3.3 V).	Translates USB to the ESP32 UART0 (TXD/RXD).
6	Q1	Dual NPN with resistors, UMH3N.	Classic auto-reset circuit: DTR/RTS from the CH340 drive IO0 and RST of the ESP32 during upload.
7	R96	470 ohm resistor.	Series resistor on the TXD line from the ESP32 to RXD of the CH340C.

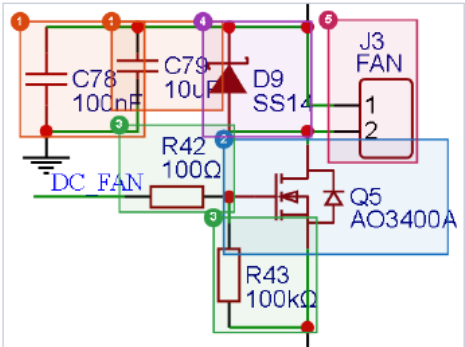
LoRa

Figure 23 - E22-900MM22S LoRa module and antenna connector



Fan driver

Figure 24 - 5 V fan driver (Q5, J3)



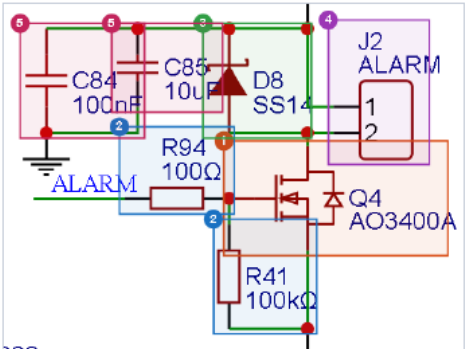
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-24	1.0	2026-07-19	5 V fan driver (Q5, J3)	24 of 26

Block function: 5 V fan driver. Requirement covered: Fan driver

No	Component	What it is	What it does
1	C79, C78	10 uF and 100 nF capacitors on +5V.	Decoupling of the +5V rail.
2	Q5	N-MOSFET AO3400Aas a low-side switch.	Turns on the fan at J3 when DC_FAN is high.
3	R42, R43	100 ohm gate resistor and 100k pull-down.	Gate current limiter and default-off holder.
4	D9	SS14 flyback diode.	Clamps the inductive spike of the fan to +5V.
5	J3	Fan connector (pin 1 = +5V, pin 2 = Q5 drain).	Connection point of the 5 V fan.

LED / buzzer driver

Figure 25 - 24 V alarm output driver (Q4, J2)



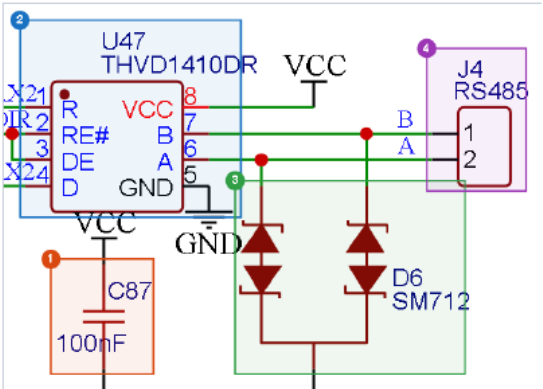
Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-25	1.0	2026-07-19	24 V alarm output driver (Q4, J2)	25 of 26

Block function: 24 V alarm output driver. Requirement covered: LED / buzzer driver

No	Component	What it is	What it does
1	Q4	N-MOSFET AO3400A as a low-side switch.	Turns on the load at J2 when ALARM is high; its source goes to GND.
2	R94, R41	100 ohm gate resistor and 100k pull-down.	R94 limits the gate charging current; R41 keeps Q4 off while the GPIO is not yet active.
3	D8	SS14 flyback diode.	Clamps the inductive spike of the load at J2 to +24V.
4	J2	Alarm load connector (pin 1 = +24V, pin 2 = Q4 drain).	Connection point of the 24 V alarm load (load type to be confirmed).
5	C84, C85	100 nF and 10 uF capacitors on +24V.	Decoupling of the +24V rail next to the alarm driver.

RS-485 / Modbus

Figure 26 - RS-485 transceiver (U47)



Drawing / document no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-SCH-26	1.0	2026-07-19	RS-485 transceiver (U47)	26 of 26

Block function: RS-485 transceiver, TVS protection and connector. **Requirement covered:** RS-485 / Modbus

No	Component	What it is	What it does
1	C87	100 nF capacitors on VCC.	Decoupling close to each 3.3 V IC.
2	U47	RS-485 transceiver THVD1410 (3.3 V).	Converts the TX2/RX2 UART of the ESP32 into the differential A/B signal; direction is controlled by DIR (drives RE# and DE together).
3	D6	TVS SM712 (asymmetric, made for RS-485).	Protects pins A/B against surges and ESD.
4	J4	RS-485 connector (A, B).	Bus connection terminal. There is no 120 ohm termination or bias resistor on this schematic.

6(a).8 PCB layout

The layout of the GLD main board is presented in ten views generated from the EasyEDA PCB file of the same project. Each view highlights one aspect required by item 6(a).8; numbered markers on the view correspond to the legend printed below it.

Board data

Parameter	Value	Source / status
Board name	Main Board GLD (MotherBoardGLDVer2)	Layout file
Board outline	Circular, Ø 84.0 mm	Dimension in view 04
Copper layers	Top and bottom (two views of copper)	Views 01, 02
Component side	All SMD parts on top; only through-hole connectors pass to the bottom	View 02
Enclosure fixing holes	4 × Ø 3.2 mm on Ø 64 mm pitch circle, at (0, +32), (+32, 0), (0, -32), (-32, 0) mm from centre	View 05
Additional holes	4 × Ø 3.0 mm near the board edge ($\approx \pm 39$ mm from centre)	View 05 — purpose (casing screw or standoff) to be confirmed against the mechanical drawing
Ground	GND pour on top and bottom; 176 of 400 vias stitch the GND pours	Views 09, 10
Laminate, thickness, copper weight	Not stated in the layout package	To be provided (FR-4 grade, UL 94 rating, Tg, CTI) — see 6.b / 6.c
Layout revision / date	Not stated in the layout package	To be added to the drawing table

Drawing register

Drawing no.	Title	Requirement covered	Rev / date
PGLD-GLD-V4-MB-PCB-01	Top layer	Top layer	TBC
PGLD-GLD-V4-MB-PCB-02	Bottom layer	Bottom layer	TBC
PGLD-GLD-V4-MB-PCB-03	Component placement	Component placement	TBC
PGLD-GLD-V4-MB-PCB-04	Board dimensions	Board dimensions	TBC
PGLD-GLD-V4-MB-PCB-05	Mounting holes	Mounting holes	TBC
PGLD-GLD-V4-MB-PCB-06	Connectors	Connectors	TBC
PGLD-GLD-V4-MB-PCB-07	High-current / power area	High-current / power area	TBC
PGLD-GLD-V4-MB-PCB-08	Sensor interface	Sensor interface	TBC
PGLD-GLD-V4-MB-PCB-09	Grounding arrangement — top	Grounding arrangement	TBC
PGLD-GLD-V4-MB-PCB-10	Grounding arrangement — bottom	Grounding arrangement	TBC

Coverage of the required layout content

Required content (6(a).8)	Where shown	Status
Top layer	View 01	Provided
Bottom layer	View 02	Provided
Component placement	View 03	Provided
Board dimensions	View 04	Provided
Mounting holes	View 05	Provided — purpose of 4 × Ø 3.0 mm holes to confirm
Connectors	View 06	Provided
High-current / power area	View 07	Provided
Sensor interface	View 08	Provided
Grounding arrangement	Views 09, 10	Provided
One set per board (Main, Sensor, Alarm, Terminal)	Main board only	Open — sensor-module board layout to be added

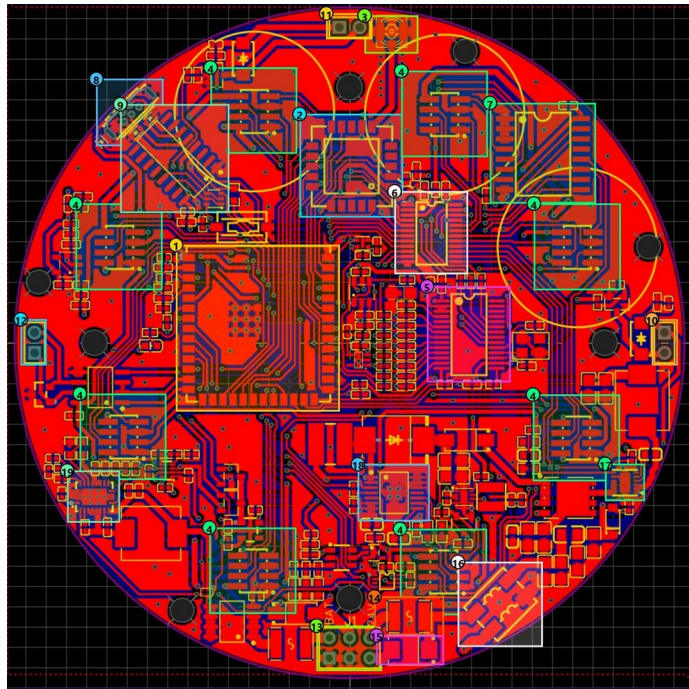


Figure 8-1. Top layer — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-01	TBC	TBC	Top layer	1 of 10

No	Item	Description
1	ESP32-S3	Main MCU module: runs the GLD firmware; SPI to LoRa and ADC, I ² C to sensors, GPIO for alarm/fan.
2	LoRa E22-900MM22S	LoRa radio for communication with the cluster head.
3	U.FL antenna connector	Connection to the external LoRa antenna.
4	Sensor ports (8)	Eight 2 × 4 headers around the board, one per sensor module (yellow circle = module outline).
5	ADS1256	24-bit ADC reading the eight analog sensor channels.
6	TCA9548A	8-channel I ² C multiplexer, one channel per sensor port.
7	PCF8574	I/O expander providing the enable signals EN0–EN7 for each sensor.
8	Micro-USB	USB port for firmware flashing and serial debug.
9	CH340C	USB-to-UART converter for programming and serial monitor.
10	ALARM connector	24 V siren/beacon output.
11	FAN connector	5 V fan output.
12	RS-485 connector	RS-485 bus (A/B).
13	VIN connector	Power input: 24V+, 24V–, 5VEXT, GND, BAT (battery).
14	PTC fuse	Limits current on the 24 V line under short circuit.
15	TVS	Clamps voltage surges on the 24 V input.
16	Common-mode filter	Filters common-mode noise on the 24 V input.
17	Buck 24 V → 5 V	LMR51450 regulator stepping 24 V down to 5 V.
18	Battery boost → 24 V	TPS61175 boosting the battery to the +24V rail.
19	Battery boost → 5 V	TPS61088 generating 5 V from the battery.

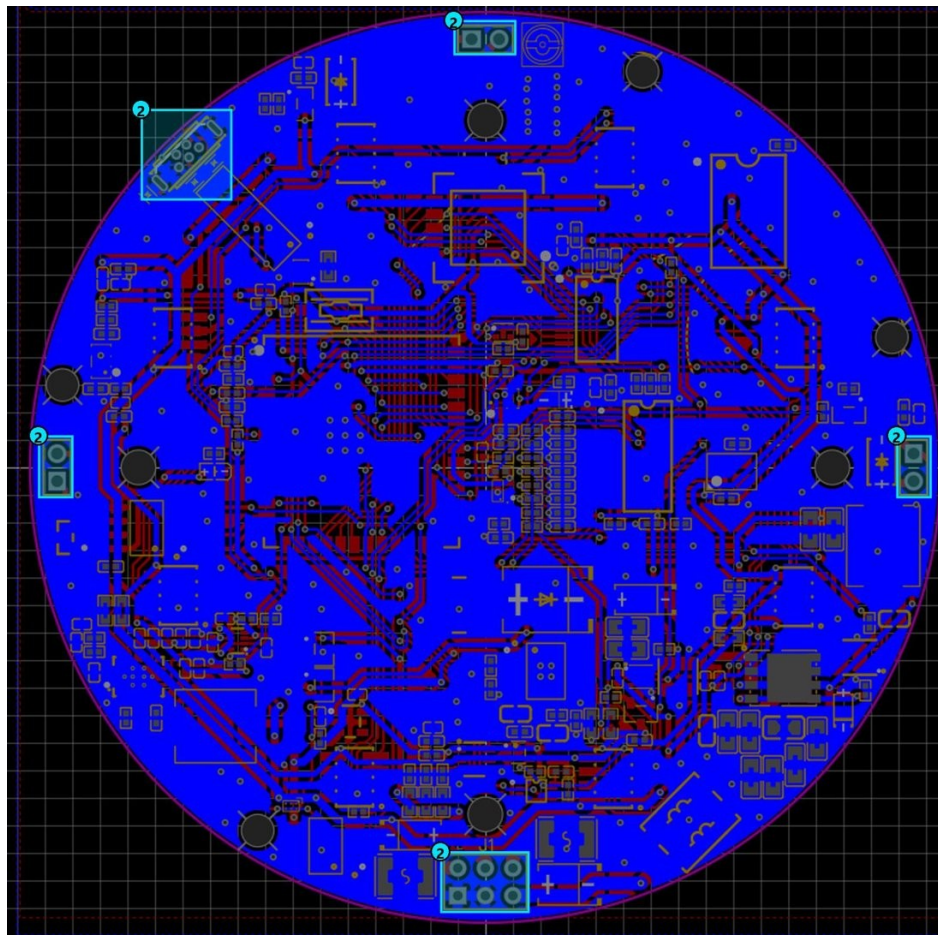


Figure 8-2. Bottom layer — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-02	TBC	TBC	Bottom layer	2 of 10

No	Item	Description
1	Bottom GND pour (solid blue area)	GND plane filling almost the whole bottom side; connected to the top pour through vias.
2	Through-hole connectors	The only parts passing through to the bottom; all SMD parts are on the top side.

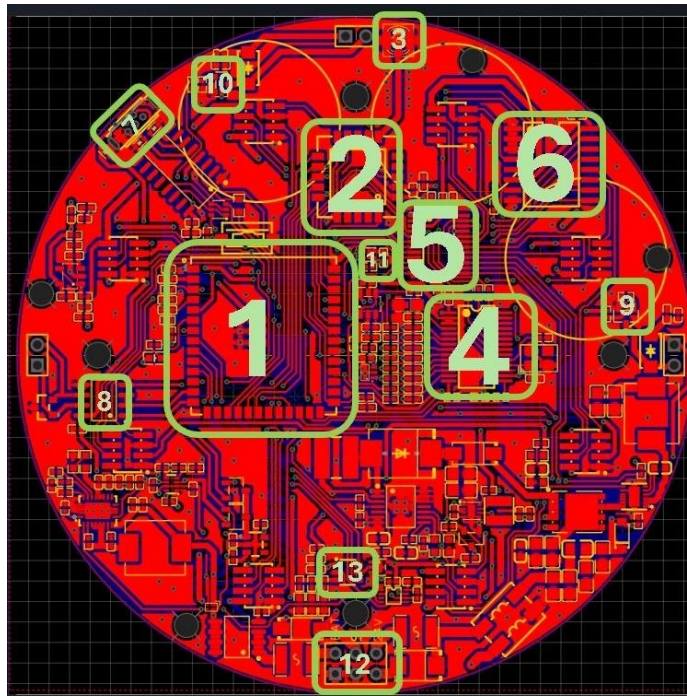


Figure 8-3. Component placement — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-03	TBC	TBC	Component placement	3 of 10

No	Item	Description
1	ESP32-S3	Main MCU: runs the firmware and controls all blocks.
2	LoRa (E22-900MM22S)	LoRa radio for communication with the cluster head.
3	Antenna (U.FL)	External LoRa antenna connector.
4	ADS1256	24-bit ADC for the eight analog sensor channels.
5	TCA9548	8-channel I ² C mux, one channel per sensor port.
6	PCF8574	I/O expander for the enable signals EN0–EN7.
7	USB	Micro-USB + CH340C for firmware flashing and serial monitor.
8	RS-485	THVD1410 transceiver for the RS-485 bus.
9	Alarm driver	MOSFET switching the 24 V siren/beacon.
10	Fan driver	MOSFET switching the 5 V fan.
11	SHT40	Temperature and humidity sensor inside the enclosure.
12	Power block	Power input and protection (VIN connector, fuse, TVS, filter).
13	TPL5010	Watchdog timer that wakes the system periodically.

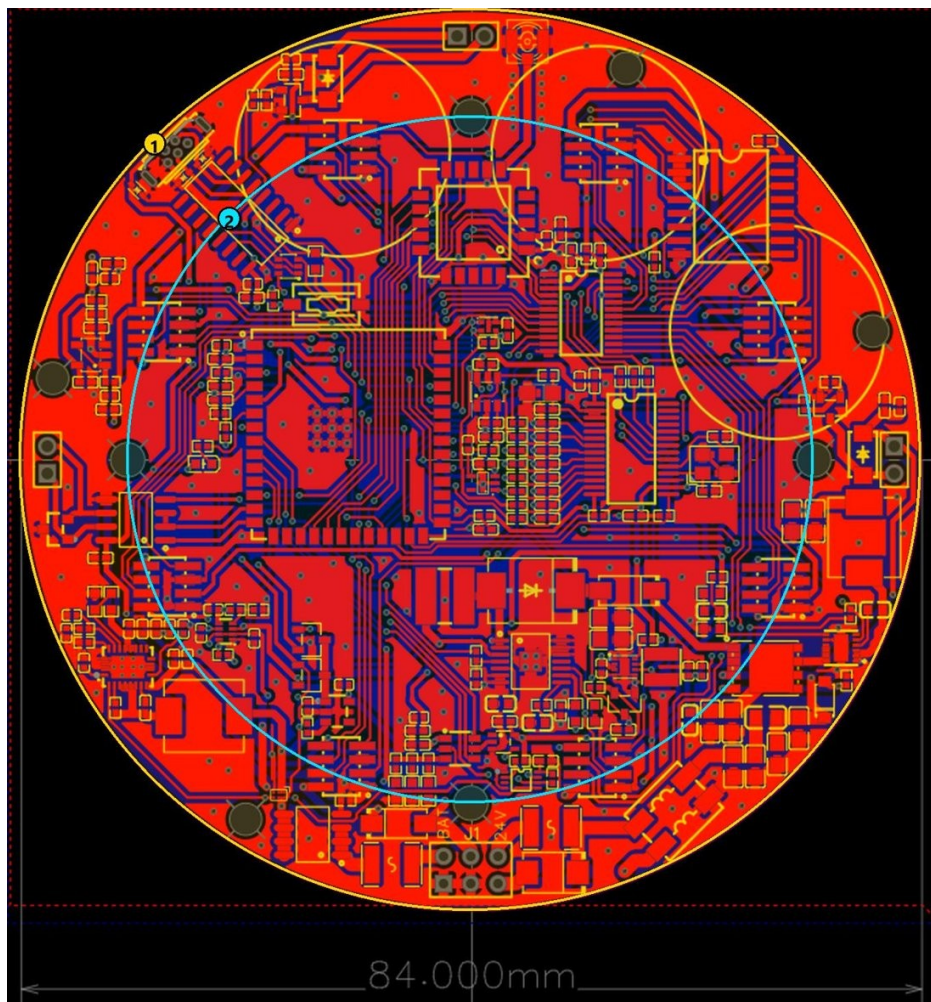


Figure 8-4. Board dimensions — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-04	TBC	TBC	Board dimensions	4 of 10

No	Item	Description
1	Board diameter: 84 mm	Circular board (outline) matching the round aluminium enclosure.
2	Enclosure-hole pitch circle: Ø 64 mm	The four enclosure fixing holes lie on this circle, 90° apart.

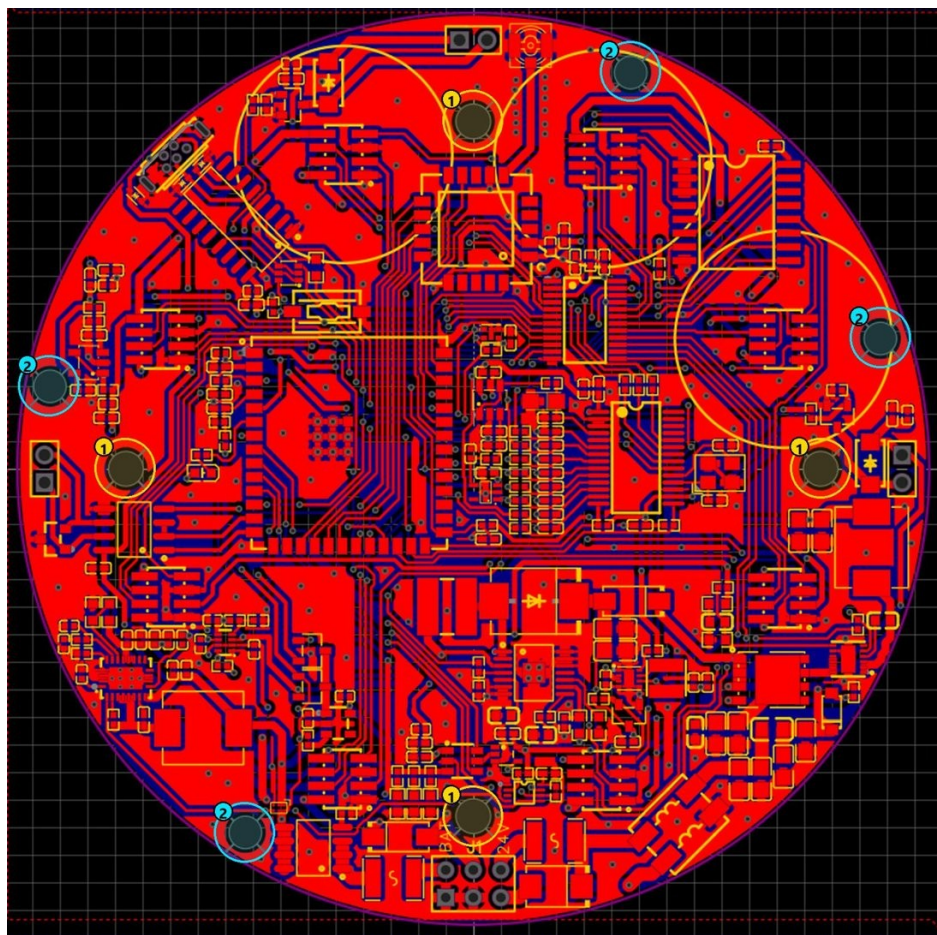


Figure 8-5. Mounting holes — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-05	TBC	TBC	Mounting holes	5 of 10

No	Item	Description
1	Enclosure holes (4 × Ø 3.2 mm)	Hole pattern of the enclosure footprint on the Ø 64 mm circle; board-to-enclosure fixing screws. Positions from centre: (0, +32), (+32, 0), (0, -32), (-32, 0) mm.
2	Additional holes (4 × Ø 3.0 mm)	Four further fixing holes near the board edge ($\approx \pm 39$ mm from centre). Exact use (enclosure screw or standoff) to be confirmed with the mechanical drawing.

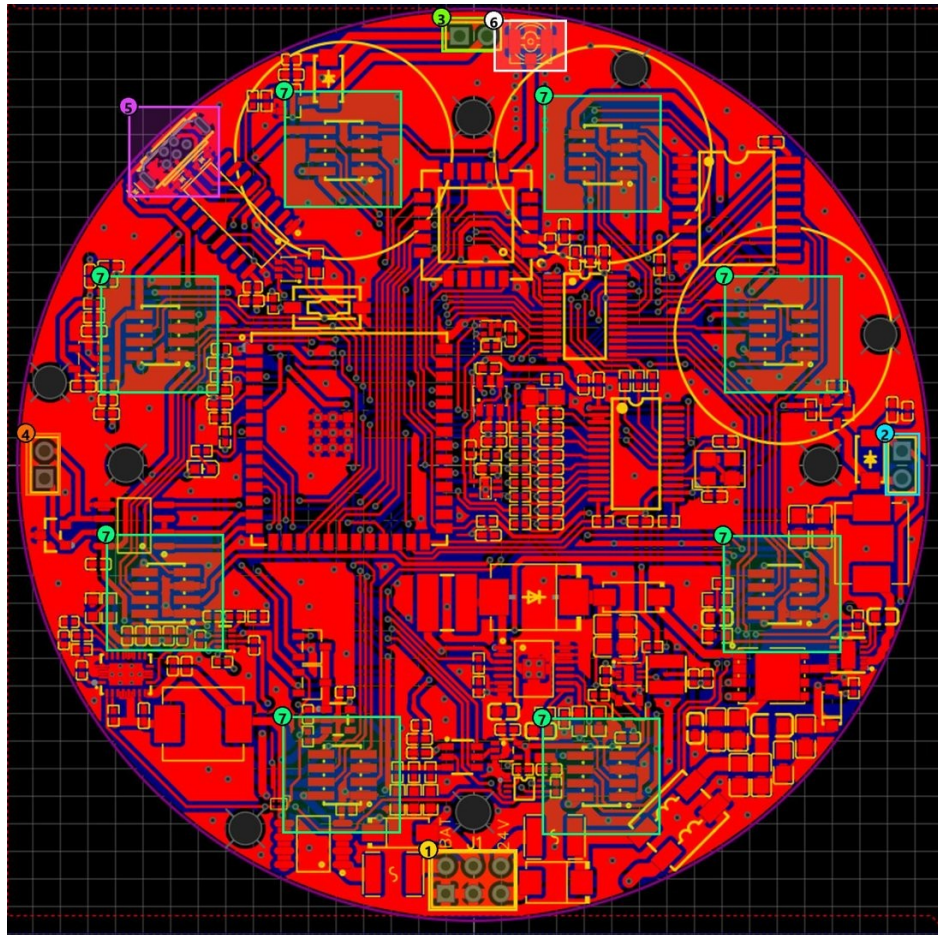


Figure 8-6. Connectors — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-06	TBC	TBC	Connectors	6 of 10

No	Item	Description
1	VIN	Power input, 2 × 3 pins: 24V+, 24V-, 5VEXT, GND, BAT (battery).
2	ALARM	24 V siren/beacon output (pin 1 = +24V, pin 2 = MOSFET switch side).
3	FAN	5 V fan output (pin 1 = +5V, pin 2 = MOSFET switch side).
4	RS-485	RS-485 bus: B (pin 1) and A (pin 2).
5	Micro-USB	Firmware flashing and serial debug.
6	U.FL (antenna)	External LoRa antenna connector.
7	Sensor ports (8)	Eight 2 × 4 headers (1.27 mm pitch): GND, +5V, AINx, SCLx, SDAX, ENx, VMID — one sensor module each.

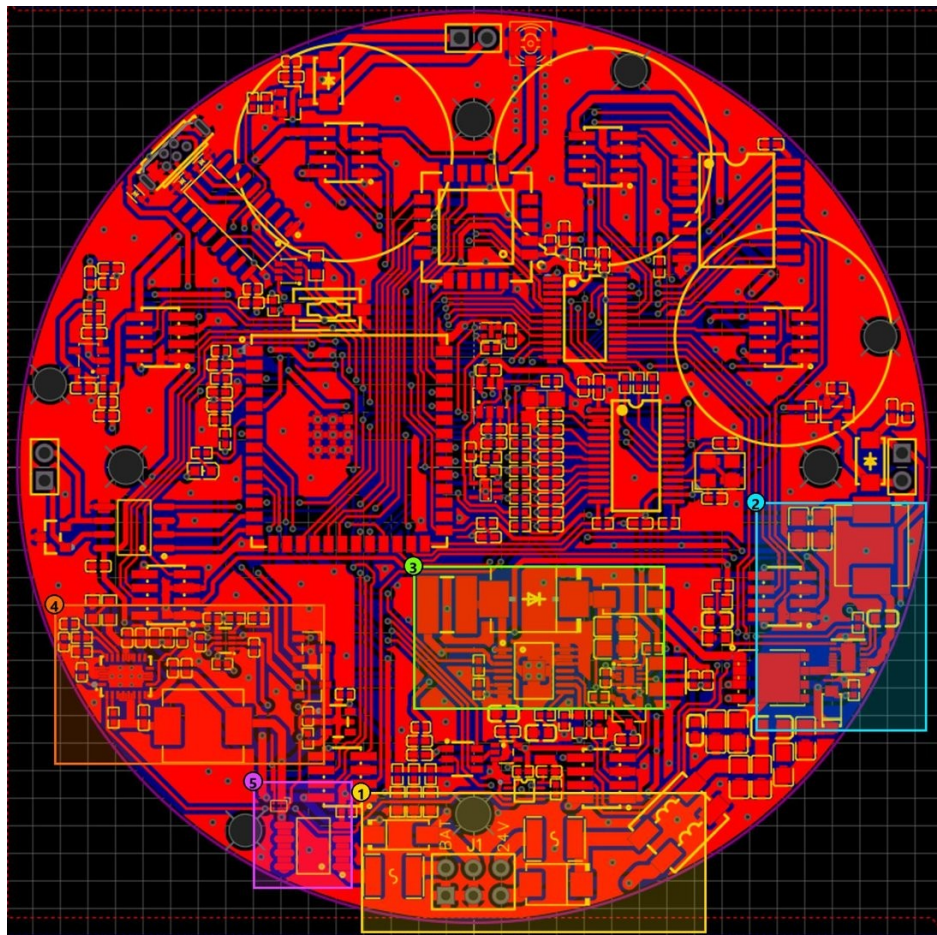


Figure 8-7. High-current / power area — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-07	TBC	TBC	High-current / power area	7 of 10

No	Item	Description
1	Power input & protection	VIN connector, fuse, TVS, common-mode filter, ferrite: 24 V entry path.
2	24 V → 5 V converter	LMR51450 buck, 4.7 μ H inductor, P-MOSFET switch.
3	Battery → 24 V converter	TPS61175 boost, 10 μ H inductor, SK36 diode and OR-ing diode to the +24V rail.
4	Battery → 5 V converter + power mux	TPS61088 boost, 2.2 μ H inductor, TPS2116 power mux selecting the 5 V source.
5	Battery protection	Fuse, P-MOSFET, load switch: disconnects the battery when the system sleeps.

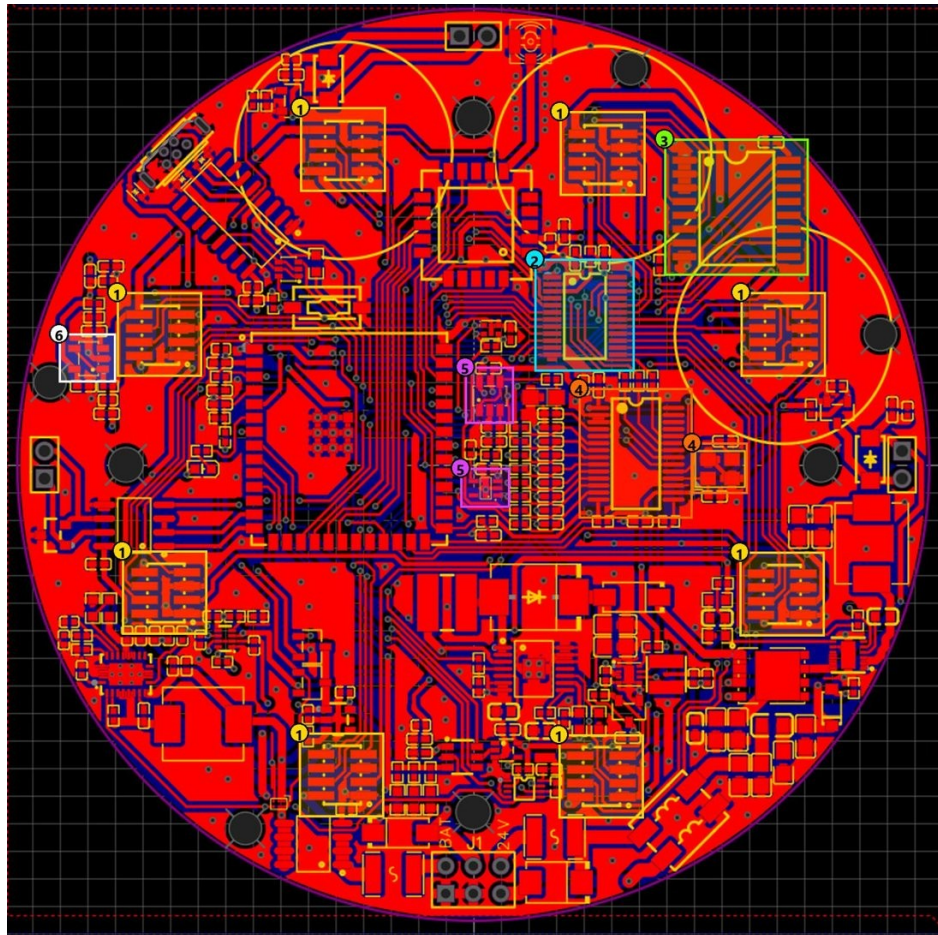


Figure 8-8. Sensor interface — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-08	TBC	TBC	Sensor interface	8 of 10

No	Item	Description
1	Sensor ports (8)	Eight headers: +5V, GND, VMID, ENx, SDAx/SCLx (I ² C per channel), AINx (analog to ADC).
2	TCA9548A	I ² C mux splitting the ESP32 I ² C bus into 8 channels for 8 sensors.
3	PCF8574	I/O expander generating EN0–EN7 to switch each sensor module.
4	ADS1256 + 8 MHz crystal	8-channel 24-bit ADC, SPI to ESP32, clocked from an 8 MHz crystal.
5	2.5 V reference + buffer	ADR03 generates VREF, buffered by OPA320 for the ADC.
6	VMID buffer	OPA320 follower generating VMID (analog mid-bias) for the sensor modules.

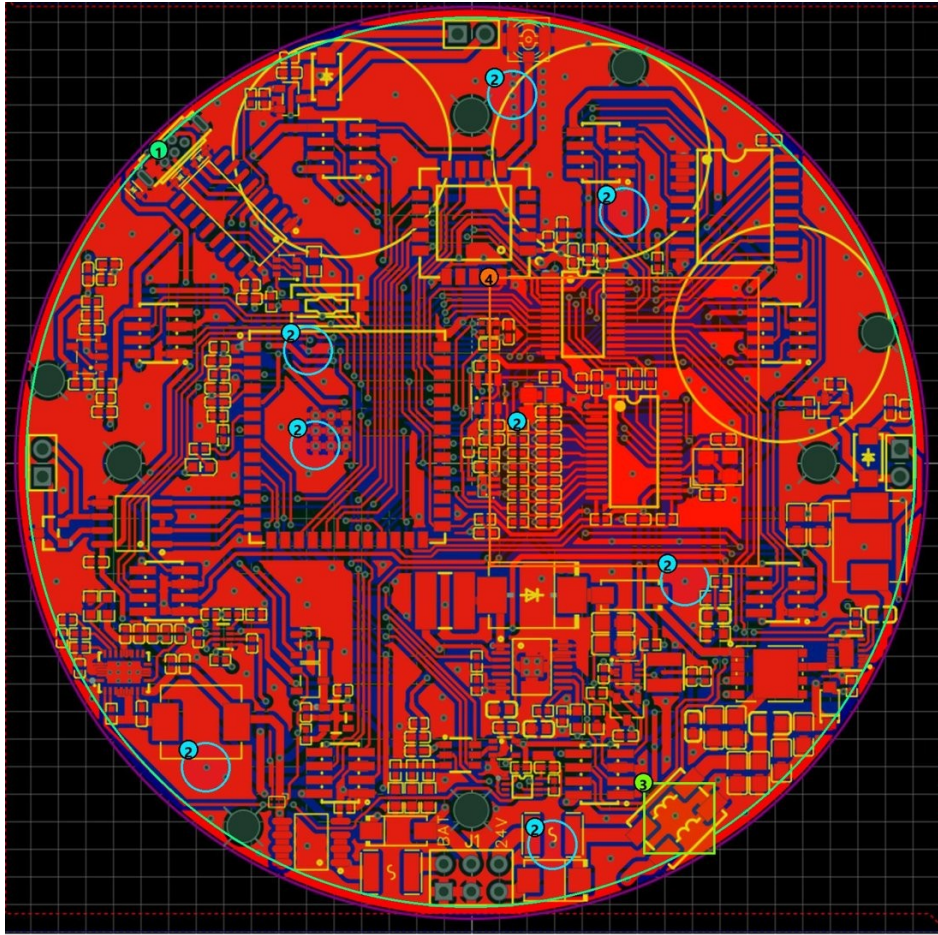


Figure 8-9. Grounding arrangement (top) — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-09	TBC	TBC	Grounding arrangement — top	9 of 10

No	Item	Description
1	GND pour	One GND copper area (green line along the board edge) on top and bottom, filling all free area: short return paths and noise shielding.
2	GND vias (examples)	176 of the 400 vias connect the top and bottom GND pours; blue circles mark examples.
3	24V- → GND at the common-mode filter	The 24 V input return (24V-) joins system GND through the return winding of the common-mode choke.
4	Analog area without split ground	Single GND net. The analog area (ADS1256, voltage reference, OPA320) is separated through the +5VA supply (ferrite), not by a split ground.

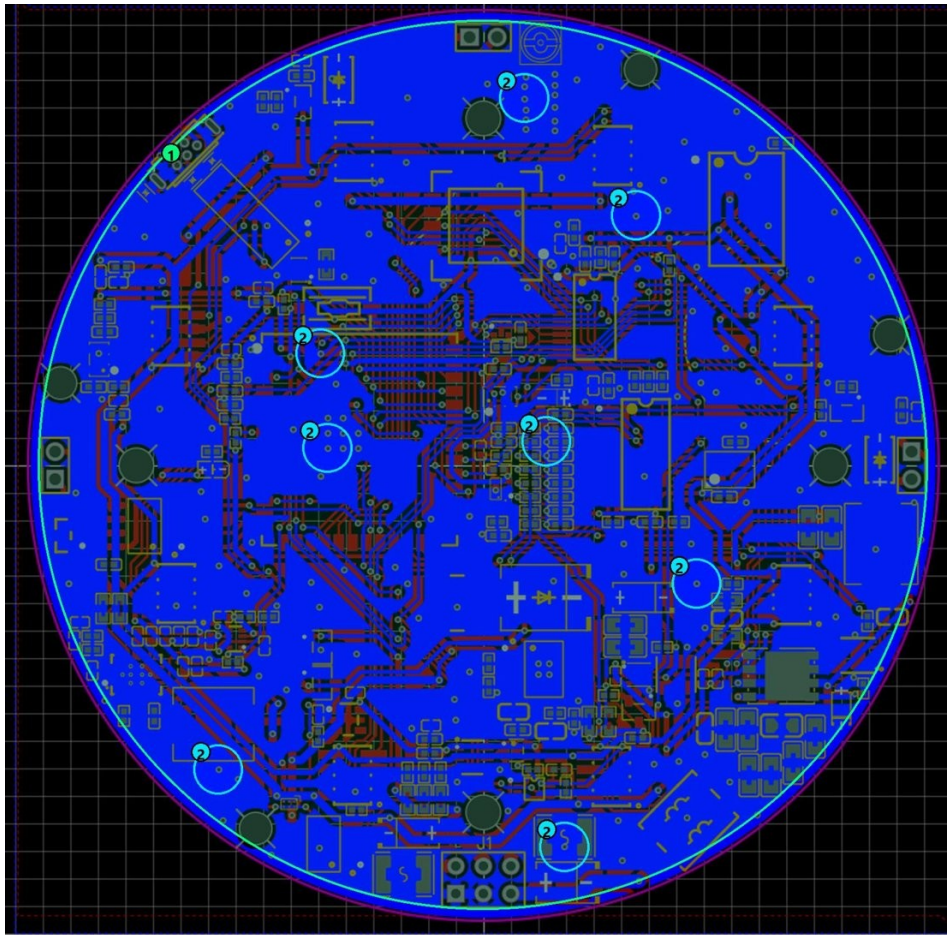


Figure 8-10. Grounding arrangement (bottom) — Main Board GLD

Drawing no.	Rev	Date	Title	Sheet
PGLD-GLD-V4-MB-PCB-10	TBC	TBC	Grounding arrangement — bottom	10 of 10

No	Item	Description
1	GND pour	Bottom GND copper area filling all free area.
2	GND vias (examples)	176 of 400 vias stitch the top and bottom GND pours; blue circles mark examples.

Open items for 6(a).8: (a) layout revision and date for the drawing table; (b) PCB laminate grade, UL 94 rating, Tg, CTI, board thickness and copper weight; (c) purpose of the 4 × Ø 3.0 mm holes; (d) layout drawings of the sensor-module board.